



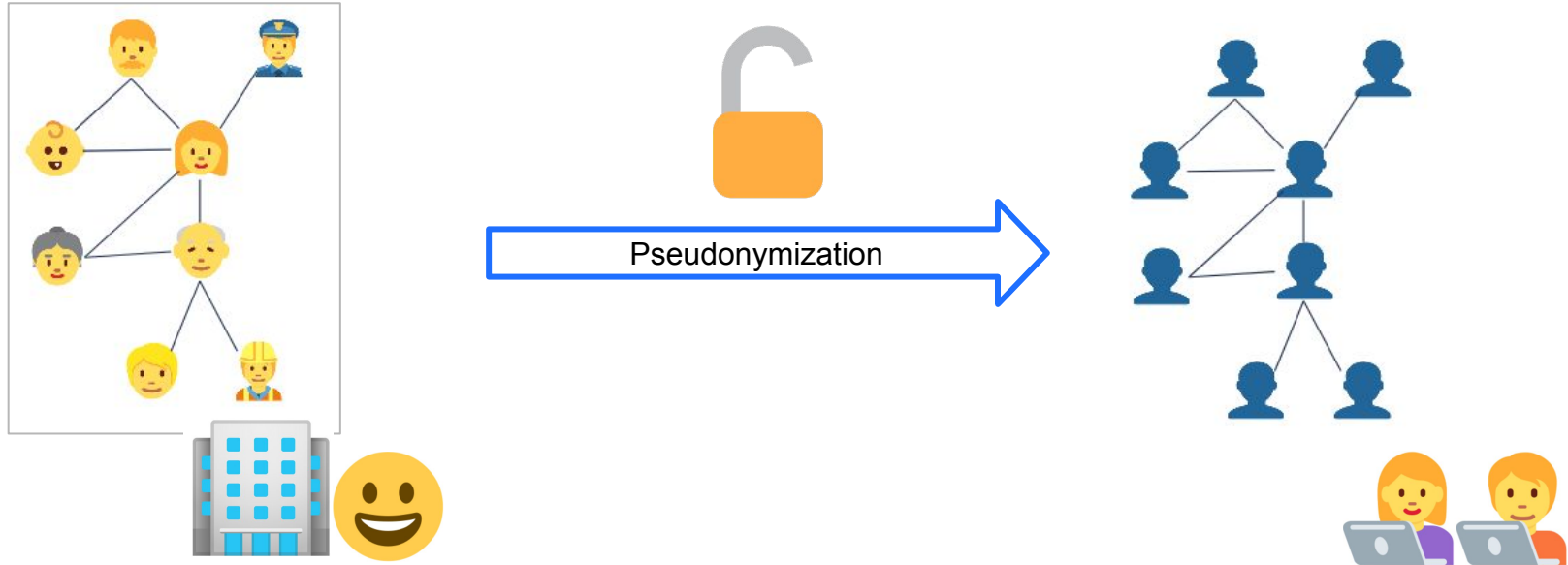
Universiteit
Leiden

Fuzzy k -anonymity in complex networks

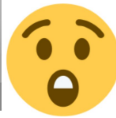
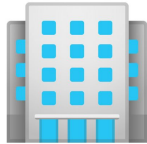
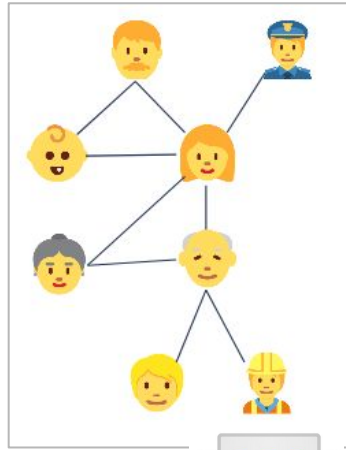
Rachel G. de Jong,
Mark P. J. van der Loo,
Prof. Frank W. Takes

Leiden University, Statistics Netherlands

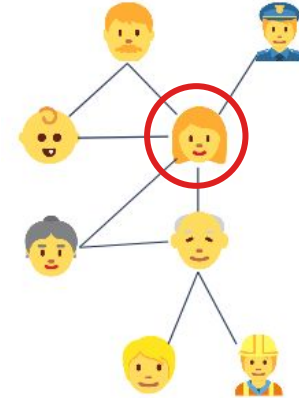
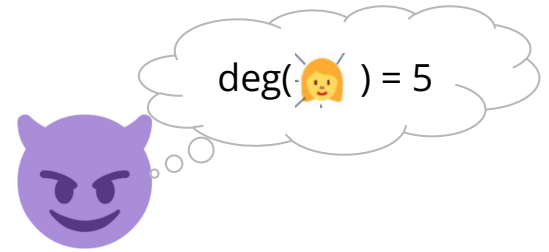
Safely publishing networks?



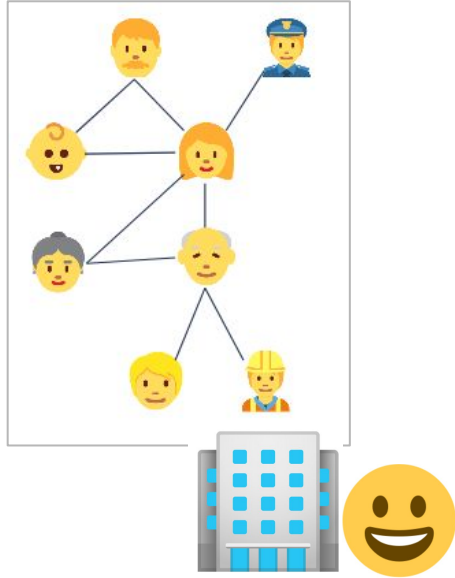
Safely publishing networks?



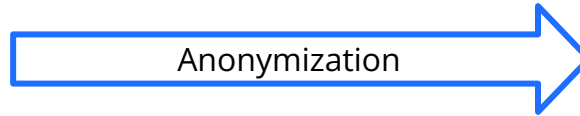
Pseudonymization



Safely publishing networks?

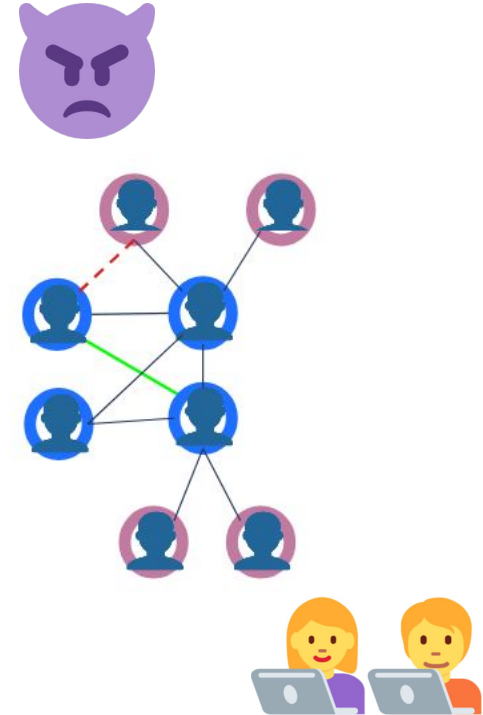


- **How to measure anonymity?**
- Preserving utility?
- Anonymization?



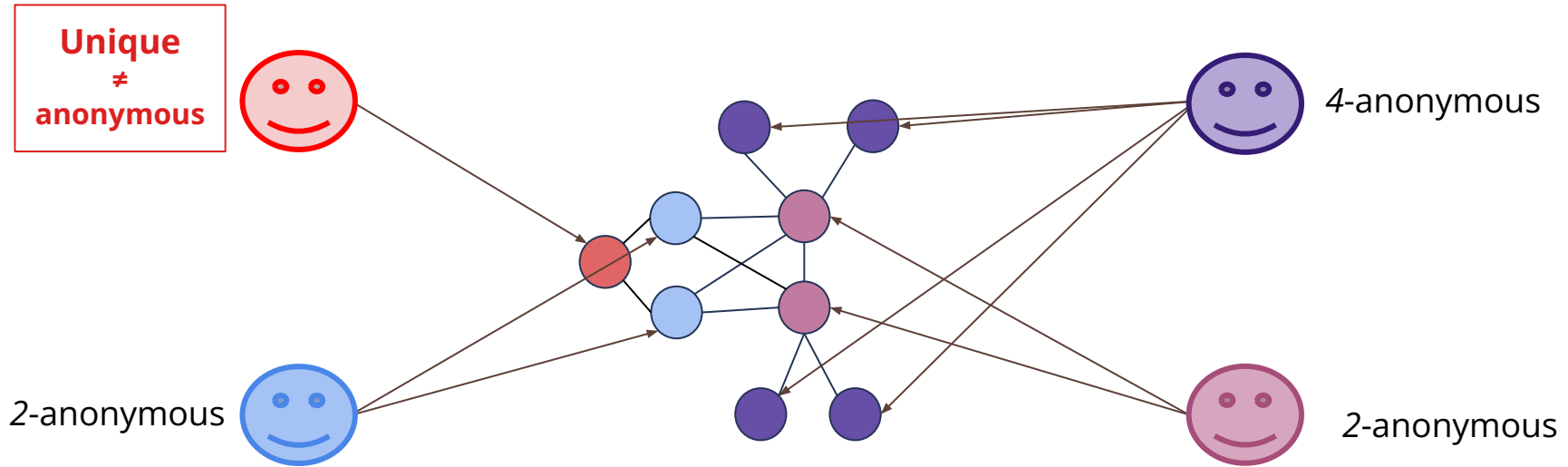
Deleting edges ---

Adding edges ---

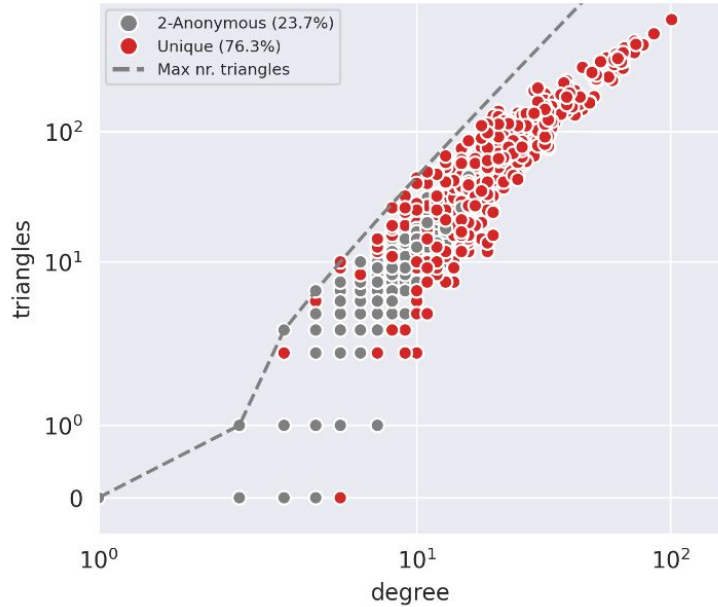


Approach: k -Anonymity

How many candidates are there for each individual?

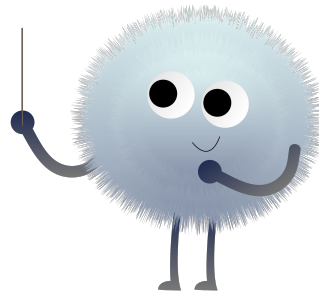


Fuzzy k -anonymity

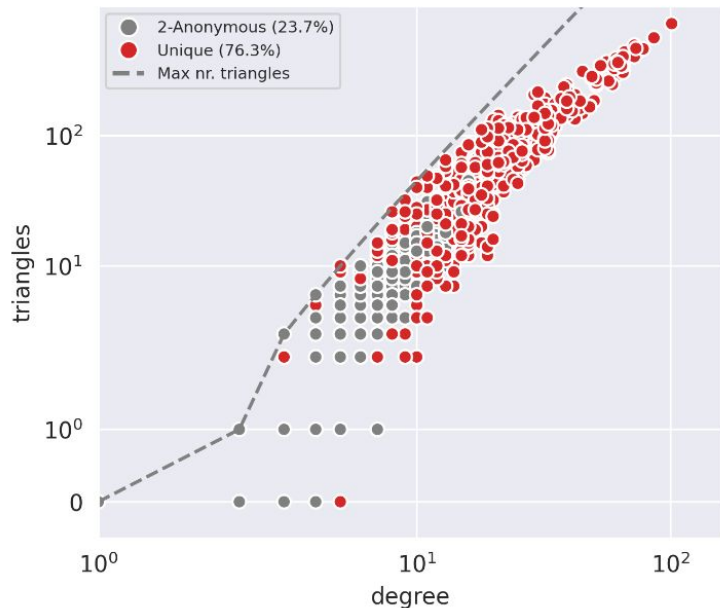


23.7% 2-anonymous

Copnet Facebook
network



Fuzzy k -anonymity

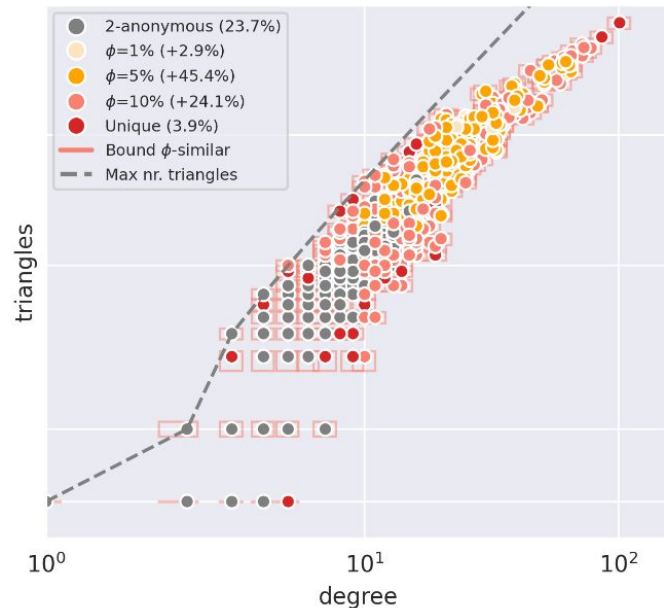
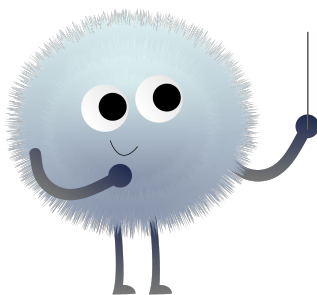


23.7% 2-anonymous

v ϕ -similar to w if:

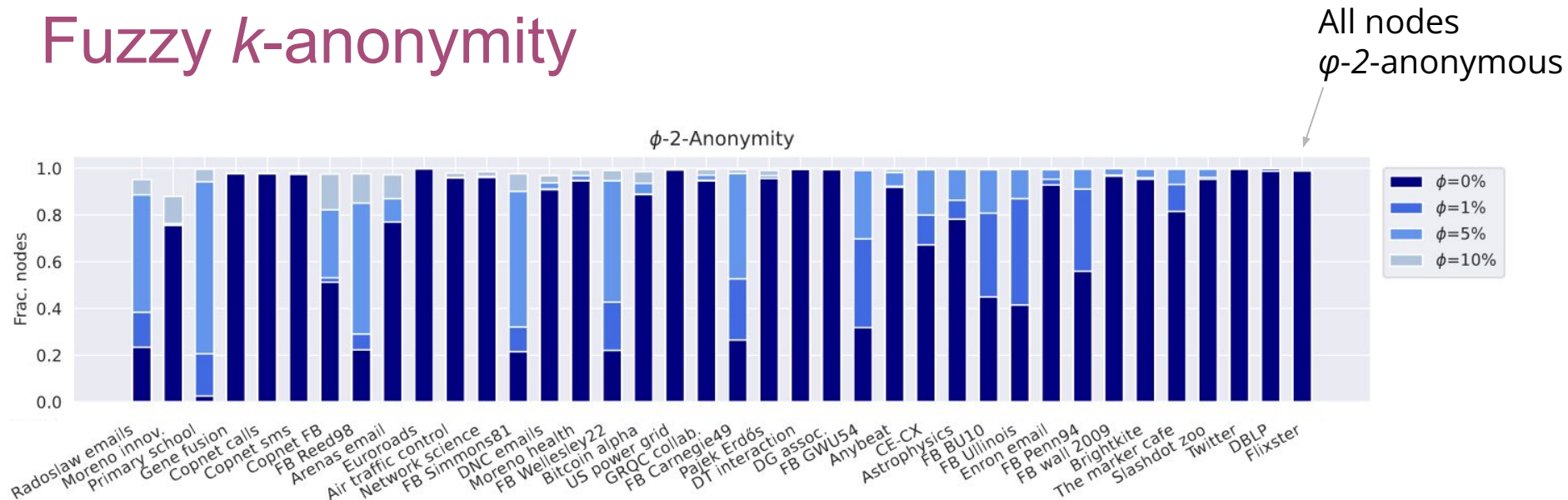
$$\frac{\Delta \deg(v, w)}{\deg(v)} \leq \phi$$

$$\frac{\Delta \text{tri}(v, w)}{\text{tri}(v)} \leq \phi$$



96.1% ϕ -2-anonymous

Fuzzy k -anonymity

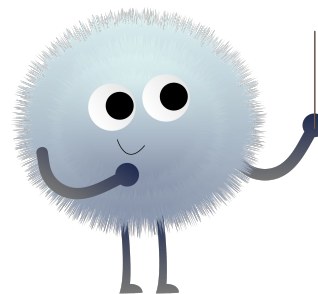


Uncertainty increases anonymity!

- $\phi = 5\% \rightarrow 64\%$ of unique nodes anonymous
- **Many unique nodes very similar to at least one other node**

Fuzzy k -anonymity

- More realistic attacker scenario
→ **Non-fuzzy overestimates risks**
- $\varphi=5\%$:
 - **Avg. + 64% unique nodes anonymous**
- $\varphi=10\%$ + greedy anonymization:
 - **Most nodes anonymous**



Read our paper!



Email

r.g.de.jong@liacs.leidenuniv.nl

 [linkedin.com/in/rachelgdejong/](https://www.linkedin.com/in/rachelgdejong/)

Website

<https://racheldejong.github.io/>